

Robert Lee

Senior Software Engineer — Search, Systems & AI

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SUMMARY

Senior Software Engineer on Microsoft Azure AI Search building retrieval systems that ground enterprise AI in governed knowledge at billion-vector scale. Experience spans vector search, agentic retrieval, information retrieval, RAG, distributed systems, graph databases, performance optimization, and production reliability.

TECHNICAL SKILLS

Languages: C++, C#, Java, Python, TypeScript

Search & AI: Vector search, vector quantization, information retrieval, hybrid search, agentic retrieval, RAG, LLM agents, graph search, computer vision

Systems: Distributed systems, durable systems, performance engineering, SIMD, authentication, benchmarking, production diagnostics

EXPERIENCE

Senior Software Engineer

Mar. 2025 – Present

Microsoft — Azure AI Search C++, C#, Java, Python

Redmond, WA

- Engineering lead for a first-of-its-kind **vector-search diversity capability**, owning scoping, design, implementation, and refinement. Adapted Microsoft Research work to broaden evidence for complex queries at **billions-of-vectors scale**.
- Led research-to-production delivery for **agentic-retrieval filter and boost generation**. Reframed an unbounded synthesis problem as a bounded, verifiable operator set, validated it against production usage, aligned research and product teams, and shipped multiple improvements to Public Preview.
- Integrated Azure AI Search into tool-calling and multi-agent workflows, grounding LLM agents in governed enterprise knowledge for Retrieval Augmented Generation.
- Built an agentic-retrieval benchmarking system across CPU, memory, throughput, latency distributions, tool iterations, and dependencies; proposed and shipped the production billing model for a new serverless enterprise search offering.
- Lead architecture reviews and technical decisions across information retrieval, vector search, Azure OpenAI, and agent orchestration while mentoring and onboarding engineers.

Software Engineer II

Jun. 2022 – Feb. 2025

Microsoft — Azure AI Search C++, C#, Java, Python

Redmond, WA

- Part of the team that scaled **vector search from 1 to N** and took **quantization from 0 to 1, then 1 to N**, building new capabilities and scaling them for broad production adoption across billions of embeddings.
- Drove **vector quantization from Public Preview through GA**, delivering **8–32x customer cost savings** and **up to 20x lower latency** through binary vectors, scalar and binary quantization, and SIMD-accelerated distance computation.
- Optimized vector distance kernels using SIMD operations, loop unrolling, multiple independent accumulators, and fused multiply-add instructions to improve instruction-level parallelism and throughput.
- Designed hybrid-search subscore fusion and score thresholding to improve relevance across blended vector and keyword retrieval.
- Designed a resource-aware quota mechanism for HNSW indexes that **reduced limit overshoot by 100x**.
- Owned an extensible facet-aggregation engine from specification through delivery, implementing a lexer, parser, and evaluator using BNF grammar, the shunting-yard algorithm, and Reverse Polish Notation.
- Root-caused cross-team production incidents, caught a critical quantization bug before release, and served as a subject-matter expert and reviewer for distributed-system and vector-algorithm changes.

Software Engineer

Jul. 2021 – May 2022

Microsoft — Azure AI Search

Redmond, WA

- Delivered index aliases to Public Preview, allowing applications to repoint to another search index with **no code changes or downtime**.
- Completed a phased, zero-downtime telemetry-table migration that preserved data integrity and produced a **50–100x query speedup**.

Software Engineer Intern

Sep. 2020 – Dec. 2020

Microsoft — Azure Cognitive Search C#

Remote

- Designed and shipped the team's **second most requested API and backend feature**, released under a preview API.

Software Engineer Intern

Jun. 2019 – Aug. 2019

Microsoft — Azure Search, AI Platform TypeScript

Bellevue, WA

- Built a dynamic search-site generator with query suggestions and filtering; improved the portal with a JSON editor and site customization.

Software Developer Intern

Jan. 2018 – Apr. 2018

Microsoft Garage C#, TensorFlow

Vancouver, BC

- Built a cross-platform mobile application using offline machine learning for chest X-ray classification, including its image pipeline, CI, iOS share extension, and TensorFlow Android integration.

SELECTED PROJECTS

Human Pose Estimation Using Deep Neural Networks

Jan. 2021 – May 2021

github.com/robertklee/COCO-Human-Pose

- Led a student team developing a heatmap-based 2D human-pose-estimation model on COCO 2017; architected the cloud training pipeline, model, data augmentation, visualization, and deployment.
- Trained the model **from scratch with random weight initialization** and built much of the supporting ML pipeline.

Semantic Road Segmentation Using Convolutional Neural Networks

Jul. 2020 – Sep. 2020

github.com/robertklee/KITTI-RoadSeg

- Trained a U-Net CNN on KITTI Road using Keras, achieving **up to 99.1% F1** and 91% in the worst case.
- Implemented the complete training pipeline, custom losses, data generation and augmentation, cloud training, evaluation, and architecture debugging.

Self-Supervised Monocular Depth Estimation

Jan. 2020 – May 2020

github.com/DeclanMcIntosh/monodepthV2tf

- Implemented a TensorFlow version of Monodepth2 using a ResNet-18 U-Net, stereo photometric reconstruction, edge-aware smoothness, multi-scale loss, auto-masking, and minimum reprojection.
- Adapted training to approximately 136,000 DrivingStereo image pairs without requiring ground-truth depth.

Battlesnake Reinforcement Learning Agent

Jan. 2019 – May 2019

- Trained a keras-rl agent through self-play and competition against public agents to make real-time survival decisions in a multi-agent Battlesnake environment.

LEADERSHIP & COMMUNITY

Industry Mentor, SENG 321 — Requirements Engineering

Aug. 2026 – Present

University of Victoria

- Authored an open-ended challenge based on unsolved production AI problems and coach a student team from ambiguous requirements through stakeholder analysis, scope, design rationale, and a clickable prototype.

Industry Mentor, SENG 321 — Requirements Engineering

Sep. 2024 – Dec. 2024

University of Victoria

Victoria, BC

- One of 18 industry mentors for a 120-student cohort; authored an LLM reliability and safety challenge and reviewed teams' requirements, technical assumptions, designs, and evaluation strategies.

Founder & Program Director, Seniors' Digital Literacy Program

Jul. 2015 – Oct. 2021

University of Victoria

Victoria, BC

- Founded and directed the program for six years, growing it to **180 volunteers, 650+ seniors served, and 30 workshops**; secured funding, lab space, sponsorship, and administrative support.
- Designed concurrent beginner and advanced curricula, built a recurring volunteer pipeline, and handed the institutionalized program to successors who sustained it beyond my tenure.

Chair & Vice-Chair, IEEE Student Branch

Sep. 2019 – Jan. 2021

University of Victoria

Victoria, BC

- Co-led a six-person executive and 70-member branch; created a **14-workshop series reaching 350+ students** across Git, machine learning, circuit design, and hardware prototyping.
- Authored and taught curriculum, secured funding and lab access, recruited industry instructors, and established a recurring program that continued after my term.

Founder & Conference Chair, Tech & Business Strategy Conference

Aug. 2019 – Nov. 2019

University of Victoria

Victoria, BC

- Secured university support and led planning, logistics, speakers, and execution for a **200+ attendee conference** connecting technology and business strategy.

EDUCATION

Bachelor of Engineering, Electrical & Computer Engineering

Sep. 2016 – Apr. 2021

University of Victoria Cumulative average: **97%**

Victoria, BC

- Coursework and projects spanning distributed systems, algorithms, signal processing, embedded design, and machine learning.
- Jamie Cassels Undergraduate Research Award recipient; researched hardware acceleration for machine-learning neural networks.

SELECTED AWARDS & CREDENTIALS

- **Y Combinator AI Startup School — Inaugural Cohort** (Jun. 2025) — Selected by application for an invitation-only, two-day gathering of approximately 2,500 AI researchers, engineers, and builders in San Francisco; **approximately 8% acceptance among 30,000 applicants**. Talks from Sam Altman, Satya Nadella, Andrej Karpathy, Fei-Fei Li, and others covered frontier models, agents, and the path from research to product.
- **Schulich Leader Scholarship** (2016) — \$80,000 full ride; one of 50 awarded nationally from approximately 1,500 school nominees.
- **National Champion, Michael Smith Science Challenge** (2014) — national-record 97.5% among 1,700+ candidates.
- **Governor General's Academic Medal, Bronze** (2016).
- **Jamie Cassels Undergraduate Research Award** (2019).
- **1st Place, Western Engineering Competition — Senior Design** (2020).

- **1st Place (3x), UVEC Engineering Competition — Senior Design (2017–2019).**